

Theme: *Supply Chain Intelligence / Energy Security / Geopolitical Risk*

PROBLEM CONTEXT

India sources approximately 88% of its crude oil from imports, with 40–45% of that volume transiting through the Strait of Hormuz. This is not a theoretical risk — it is a structural vulnerability that geopolitical events repeatedly stress-test. The 2025 US-Iran standoff sent Brent crude up over 8% in a single session and forced Indian refiners onto spot markets at steep premiums. The situation has not stabilised: through early 2026, renewed US sanctions pressure on Iranian oil exports, maritime security incidents in the Persian Gulf, and escalating Houthi attacks on Red Sea shipping lanes have kept the supply threat live. India's Strategic Petroleum Reserves provide roughly 9.5 days of national consumption cover — a buffer that would be exhausted quickly in any sustained disruption. A McKinsey analysis of past energy supply shocks found that economies without automated rerouting and demand-management capability took an average of 47 days longer to stabilise supply than those with integrated response intelligence. Traditional supply chain planning tools were designed for predictable environments. They have no ability to model geopolitical scenario impacts in real time, dynamically evaluate alternative procurement corridors, or orchestrate coordinated response across refiners, logistics providers, and strategic reserves. That intelligence layer needs to be built.

CHALLENGE STATEMENT

Design an AI-powered Energy Supply Chain Resilience system that monitors geopolitical and logistics risk signals continuously, models disruption scenarios and their downstream economic impacts, and generates executable procurement rerouting recommendations — turning a reactive crisis response into a managed, anticipatory process.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Geopolitical Risk Intelligence Agent** — Multi-source agent that ingests news feeds, shipping AIS data, sanctions registries, and commodity price signals to produce a live supply disruption probability score by corridor and supplier — updated continuously, not weekly.
- **Disruption Scenario Modeller** — AI system that simulates specific events — Hormuz partial closure, OPEC+ emergency cut, Red Sea shipping suspension — and computes cascading impacts on refinery run rates, domestic fuel prices, power sector stress, and GDP trajectory under each scenario.
- **Adaptive Procurement Orchestrator** — Agentic system that identifies and ranks alternative crude sources and logistics routes, factoring in spot market pricing, tanker availability, port congestion, and refinery grade compatibility — generating procurement recommendations that procurement teams can act on within hours.
- **Strategic Reserve Optimisation Agent** — AI layer that models optimal SPR drawdown schedules against supply gap forecasts, refinery demand curves, and replenishment window estimates — providing decision support to policymakers under time pressure.
- **Supply Chain Digital Twin** — Geospatial simulation of India's full energy supply network — from wellhead to refinery to distribution — that enables continuous 'what-if' analysis and serves as a persistent intelligence platform for supply chain resilience planning.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Agentic AI / Multi-Agent Systems
- Geospatial Intelligence (AIS vessel tracking, pipeline and port mapping)
- Predictive Analytics & Scenario Simulation
- RAG over geopolitical and commodity intelligence sources
- Knowledge Graphs (supplier-route-risk-refinery relationships)
- LLMs for policy and news signal extraction

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Disruption signal detection lead time and accuracy, quality and executability of procurement alternatives generated, scenario model fidelity (assumptions must be explicit and testable), geospatial evidence depth, and demonstrated end-to-end response time from signal to recommendation.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%